

GC×IMS Compound Matching

How `drift_relative` is determined, and how a peak is matched to a compound from `drift-rel + RI`

Source modules: `rip.py`, `match.py` · GC-IMS-PEAK Identify workflow, Stage 1 + Stage 5

The key idea: both sides compare the same dimensionless ratio to the Reactant Ion Peak (RIP) — never an absolute drift time. That is what lets IMS matching run without any KO calibration.

1. Determining the peak's `drift_relative`

Produced in Stage 1 (`rip.py`). First locate the RIP on the drift axis: take the `RT = 0` row of the intensity matrix, skip the first 200 samples, and take the position of the maximum.

$$r = \operatorname{argmax}_{j \geq 200} I[0, j]$$

Then each peak's own drift index j_p (its `dt_index`) is divided by that RIP index to give a relative value:

$$d_p = \frac{j_p}{r}$$

The RIP itself is exactly 1.0; a value $d_p = 1.13$ means the peak drifted 13% farther than the RIP. Being a pure ratio, it is immune to sample rate, tube length, temperature and pressure.

2. The library data — and the unit filter

Each `.iml` library row stores its drift value in `Dt[a.u.]`, but the unit convention varies and is tagged by `DtMode`. The same compound can appear under different conventions:

DtMode	What Dt[a.u.] means	Comparable to d_p ?
RIPrel	Ratio to the RIP (library's own j_ℓ / r_ℓ)	Yes — same dimensionless ratio
1/K0	Inverse reduced mobility	No — needs KO calibration
ms	Raw drift time (milliseconds)	No — different unit

So `match_drift_rel()`'s first gate keeps only `DtMode == "RIPrel"` rows — the ones in the same unit as the peak. Everything else is skipped; mixing units would produce meaningless matches.

3. The drift-axis match condition

For each surviving RIPrel row ℓ , compute the absolute difference between the library drift ratio d_ℓ and the peak's d_p , and test it against the tolerance half-width τ_d :

$$\Delta_d(\ell) = |d_\ell - d_p| \leq \tau_d = 0.05$$

Hits are sorted ascending by Δ_d ; the closest one fills the IMS column, e.g. 1.139 ($\Delta 0.026$).

4. Two axes intersect → the confident identification

Drift agreement alone only says “something drifts nearby.” The identification comes from intersecting with the GC/RI axis, which uses the same shape of test:

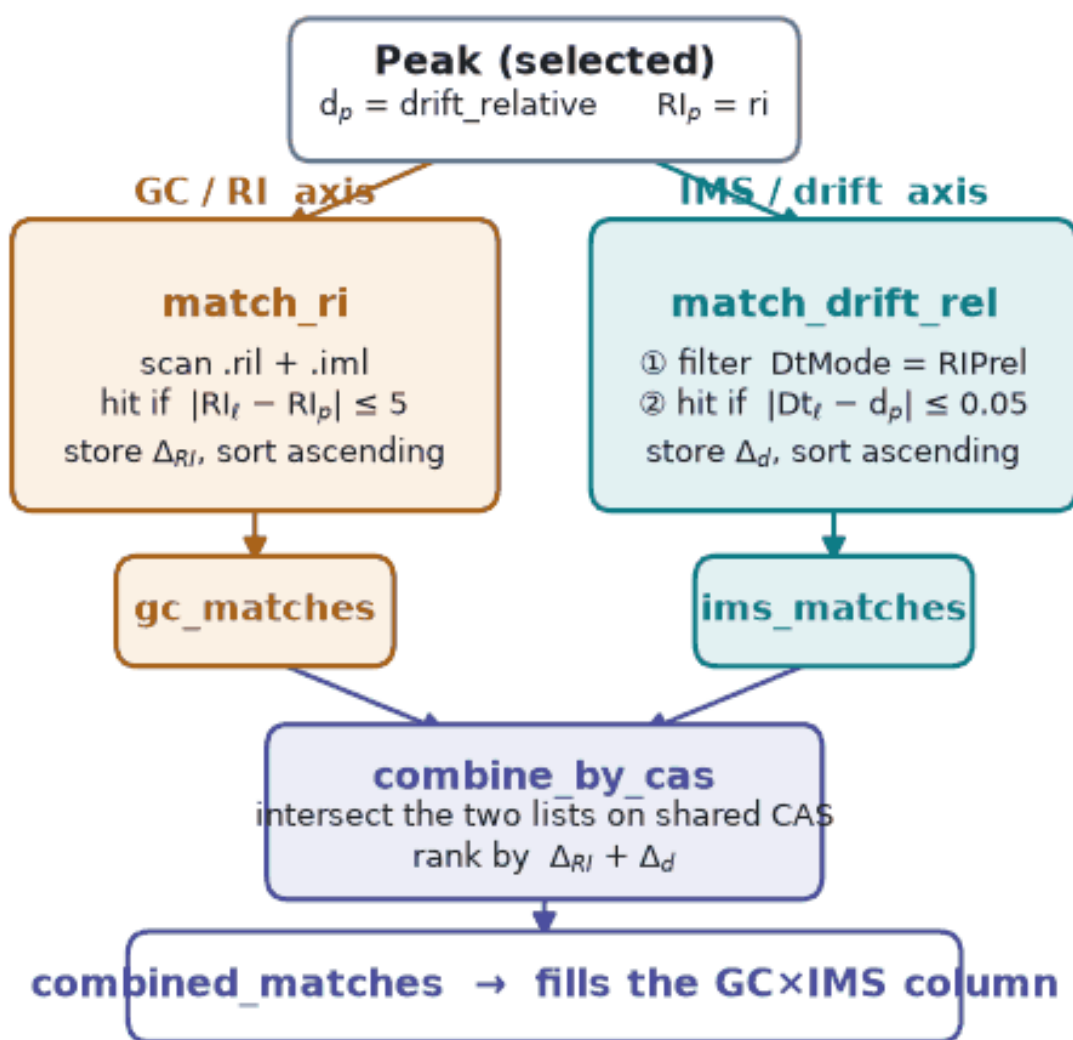
$$\Delta_{RI}(\ell) = |RI_\ell - RI_p| \leq \tau_{RI} = 5$$

`combine_by_cas()` then keys on the CAS number and keeps only compounds whose CAS lands inside both windows:

$$combined = \{c : (\Delta_{RI} \leq \tau_{RI}, \quad CAS = c) \wedge (\Delta_d \leq \tau_d, \quad CAS = c)\}$$

The intersection is ranked by the combined error $\Delta_{RI} + \Delta_d$ and shown in the GC×IMS column. Agreement on both axes collapses the RI axis’s hundreds of single-axis candidates down to a few.

5. End-to-end flow (match_all for one peak)



Amber = GC/RI axis · Teal = IMS/drift axis · Indigo = both. The KO branch is currently dormant.

Current status

The KO branch is dormant (no calibration standard yet), so the IMS axis runs on match_drift_rel (RIPrel). Both windows, $\tau_d = 0.05$ and $\tau_{RI} = 5$, are still uncalibrated placeholders and should be validated against known compounds on a reference run.